

Chapter 7.5

Integrating Rational Functions by Partial Fractions

Introduction

Every proper rational function can be expressed as a sum

$$\frac{P(x)}{Q(x)} = F_1(x) + F_2(x) + \cdots + F_n(x)$$

where $F_1(x), F_2(x), \dots, F_n(x)$ are rational functions of the form

$$\frac{A}{(ax+b)^k} \quad \text{or} \quad \frac{Ax+B}{(ax^2+bx+c)^k}$$

in which the denominators are factors of $Q(x)$. The sum is called the **partial fraction decomposition** of

$\frac{P(x)}{Q(x)}$ and the terms are called **partial fractions**.

Finding The Form of a Partial Fraction Decomposition

Linear Factor Rule

For each factor of the form

$$(ax+b)^m$$

the partial fraction decomposition contains the following sum of m partial fractions:

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_m}{(ax+b)^m}$$

where $A_1, A_2, \dots, A_m$ are constants to be determined.

Example 1 Evaluate $\int \frac{dx}{x^2+x-2}$.

Solution: We can write

$$\frac{1}{x^2+x-2} = \frac{1}{(x+2)(x-1)} = \frac{-1/3}{x+2} + \frac{1/3}{x-1}$$

$$\begin{aligned} \text{Hence, } \int \frac{dx}{x^2+x-2} &= \frac{1}{3} \int \frac{1}{x-1} dx - \frac{1}{3} \int \frac{1}{x+2} dx \\ &= \frac{1}{3} \ln|x-1| - \frac{1}{3} \ln|x+2| + c \\ &= \frac{1}{3} \ln \left| \frac{x-1}{x+2} \right| + c \end{aligned}$$

Example 2 Evaluate $\int \frac{2x+4}{x^3-2x^2} dx$.

Solution: We can write

$$\frac{2x+4}{x^3-2x^2} = \frac{2x+4}{x^2(x-2)} = \frac{-2}{x} + \frac{-2}{x^2} + \frac{2}{x-2}$$

$$\begin{aligned} \text{Hence, } \int \frac{2x+4}{x^3-2x^2} dx &= -2 \int \frac{1}{x} dx - 2 \int \frac{1}{x^2} dx + 2 \int \frac{1}{x-2} dx \\ &= -2 \ln|x| + \frac{2}{x} + 2 \ln|x-2| + c \\ &= 2 \ln \left| \frac{x-2}{x} \right| + \frac{2}{x} + c \end{aligned}$$

Quadratic Factor Rule

For each factor of the form

$$(ax^2 + bx + c)^m$$

the partial fraction decomposition contains the following sum of m partial fractions:

$$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \dots + \frac{A_mx+B_m}{(ax^2+bx+c)^m}$$

where $A_1, A_2, \dots, A_m; B_1, B_2, \dots, B_m$ are constants to be determined.

Example 3 Evaluate $\int \frac{x^2+x-2}{3x^3-x^2+3x-1} dx$.

Solution: We can write

$$\begin{aligned} \frac{x^2+x-2}{3x^3-x^2+3x-1} &= \frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{A}{3x-1} + \frac{Bx+C}{x^2+1} \\ \Rightarrow x^2+x-2 &= A(x^2+1) + (Bx+C)(3x-1) \\ \Rightarrow A &= -\frac{7}{5}, B = \frac{4}{5} \text{ and } C = \frac{3}{5} \end{aligned}$$

$$\begin{aligned} \text{Hence, } \int \frac{x^2+x-2}{3x^3-x^2+3x-1} dx &= -\frac{7}{5} \int \frac{1}{3x-1} dx + \frac{2}{5} \int \frac{2x}{x^2+1} dx + \frac{3}{5} \int \frac{1}{x^2+1} dx \\ &= -\frac{7}{5} \times \frac{1}{3} \ln|3x-1| + \frac{2}{5} \ln(x^2+1) + \frac{3}{5} \tan^{-1} x + c \\ &= -\frac{7}{15} \ln|3x-1| + \frac{2}{5} \ln(x^2+1) + \frac{3}{5} \tan^{-1} x + c \end{aligned}$$

Example 4 Evaluate $\int \frac{3x^4+4x^3+16x^2+20x+9}{(x+2)(x^2+3)^2} dx$.

Solution: We can write

$$\frac{3x^4+4x^3+16x^2+20x+9}{(x+2)(x^2+3)^2} = \frac{A}{x+2} + \frac{Bx+C}{x^2+3} + \frac{Dx+E}{(x^2+3)^2}$$

$$\begin{aligned} \Rightarrow 3x^4+4x^3+16x^2+20x+9 &= A(x^2+3)^2 + (Bx+C)(x+2)(x^2+3) + (Dx+E)(x+2) \\ &= (A+B)x^4 + (2B+C)x^3 + (6A+3B+2C+D)x^2 + (6B+3C+2D+E)x + (9A+6C+2E) \end{aligned}$$

$$\Rightarrow A+B=3, \quad 2B+C=4, \quad 6A+3B+2C+D=16, \quad 6B+3C+2D+E=20, \quad 9A+6C+2E=9$$

If $x = -2$, then we get

$$A=1, B=2, C=0, D=4, E=0$$

$$\begin{aligned} \text{Hence, } \int \frac{3x^4 + 4x^3 + 16x^2 + 20x + 9}{(x+2)(x^2+3)^2} dx &= \int \frac{1}{x+2} dx + \int \frac{2x}{x^2+3} dx + \int \frac{4x}{(x^2+3)^2} dx \\ &= \ln|x+2| + \ln(x^2+3) - \frac{2}{x^2+3} + c \end{aligned}$$

Integrating Improper Rational Functions

Example 5 Evaluate $\int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} dx$.

Solution: Using Long Division, we can write

$$\frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} = (3x^2 + 1) + \frac{1}{x^2 + x - 2}$$

$$\text{Now } \frac{1}{x^2 + x - 2} = \frac{A}{x-1} + \frac{B}{x+2} = \frac{1/3}{x-1} + \frac{-1/3}{x+2}$$

$$\begin{aligned} \text{Hence, } \int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} dx &= \int (3x^2 + 1) dx + \frac{1}{3} \int \frac{1}{x-1} dx - \frac{1}{3} \int \frac{1}{x+2} dx \\ &= x^3 + x + \frac{1}{3} \left| \frac{x-1}{x+2} \right| + c \end{aligned}$$

EXERCISE SET 7.5

9. Evaluate $\int \frac{dx}{x^2 - 3x - 4}$.

Solution: We can write

$$\frac{1}{x^2 - 3x - 4} = \frac{1/5}{x-4} + \frac{-1/5}{x+1}$$

$$\begin{aligned} \text{Therefore, } \int \frac{dx}{x^2 - 3x - 4} &= \frac{1}{5} \int \frac{1}{x-4} dx - \frac{1}{5} \int \frac{1}{x+1} dx \\ &= \frac{1}{5} \ln \left| \frac{x-4}{x+1} \right| + c \end{aligned}$$

12. Evaluate $\int \frac{5x-5}{3x^2-8x-3} dx$.

Solution: We can write

$$\frac{5x-5}{3x^2-8x-3} = \frac{1}{x-3} + \frac{2}{3x+1}$$

$$\begin{aligned} \text{Therefore, } I &= \int \frac{5x-5}{3x^2-8x-3} dx = \int \frac{1}{x-3} dx + 2 \int \frac{1}{3x+1} dx \\ &= \ln|x-3| + \frac{2}{3} \ln|3x+1| + c \end{aligned}$$

14. Evaluate $\int \frac{dx}{x(x^2-1)}$.

Solution: We can write

$$\frac{1}{x(x^2-1)} = -\frac{1}{x} + \frac{1/2}{x+1} + \frac{1/2}{x-1}$$

$$\begin{aligned}\text{Therefore, } \int \frac{dx}{x(x^2-1)} &= -\int \frac{1}{x} dx + \frac{1}{2} \int \frac{1}{x+1} dx + \frac{1}{2} \int \frac{1}{x-1} dx \\ &= -\ln|x| + \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| + c\end{aligned}$$

16. Evaluate $\int \frac{x^2+1}{x-1} dx$.

Solution: We can write

$$\frac{x^2+1}{x-1} = \frac{(x^2-1)+2}{x-1} = (x+1) + \frac{2}{x-1}$$

$$\begin{aligned}\text{Therefore, } \int \frac{x^2+1}{x-1} dx &= \int (x+1) dx + 2 \int \frac{1}{x-1} dx \\ &= \frac{x^2}{2} + x + 2 \ln|x-1| + c\end{aligned}$$

17. Evaluate $\int \frac{3x^2-10}{x^2-4x+4} dx$.

Solution: We can write

$$\frac{3x^2-10}{x^2-4x+4} = \frac{3(x^2-4x+4)+12x-22}{(x-2)^2} = 3 + \frac{12x-22}{(x-2)^2}$$

$$\text{Now } \frac{12x-22}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2} = \frac{12}{x-2} + \frac{2}{(x-2)^2}$$

$$\begin{aligned}\text{Therefore, } \int \frac{3x^2-10}{x^2-4x+4} dx &= 3 \int dx + 12 \int \frac{1}{x-2} dx + 2 \int \frac{1}{(x-2)^2} dx \\ &= 3x + 12 \ln|x-2| - \frac{2}{x-2} + c\end{aligned}$$

21. Evaluate $\int \frac{x^5+x^2+2}{x^3-x} dx$.

Solution: We can write

$$\begin{aligned}\frac{x^5+x^2+2}{x^3-x} &= (x^2+1) + \frac{x^2+x+2}{x^3-x} = (x^2+1) + \frac{x^2+x+2}{x(x+1)(x-1)} \\ &= (x^2+1) + \frac{0^2+0+2}{x(0+1)(0-1)} + \frac{(-1)^2-1+2}{(-1)(x+1)(-1-1)} + \frac{1^2+1+2}{(1)(1+1)(x-1)} \\ &= (x^2+1) + \frac{-2}{x} + \frac{1}{x+1} + \frac{2}{x-1}\end{aligned}$$

Therefore,

$$\int \frac{x^5+x^2+2}{x^3-x} dx = \int (x^2+1) dx - 2 \int \frac{1}{x} dx + \int \frac{1}{x+1} dx + 2 \int \frac{1}{x-1} dx = \frac{x^3}{3} + x - 2 \ln|x| + \ln|x+1| + 2 \ln|x-1| + c$$

22. Evaluate $\int \frac{x^5 - 4x^3 + 1}{x^3 - 4x} dx$.

Solution: We can write

$$\begin{aligned}\frac{x^5 - 4x^3 + 1}{x^3 - 4x} &= \frac{x^2(x^3 - 4x) + 1}{x^3 - 4x} = x^2 + \frac{1}{x(x+2)(x-2)} \\ &= x^2 + \frac{-1/4}{x} + \frac{1/8}{x+2} + \frac{1/8}{x-2}\end{aligned}$$

$$\begin{aligned}\text{Therefore, } \int \frac{x^5 - 4x^3 + 1}{x^3 - 4x} dx &= \int x^2 dx - \frac{1}{4} \int \frac{1}{x} dx + \frac{1}{8} \int \frac{1}{x+2} dx + \frac{1}{8} \int \frac{1}{x-2} dx \\ &= \frac{x^3}{3} - \frac{1}{4} \ln|x| + \frac{1}{8} \ln|x+2| + \frac{1}{8} \ln|x-2| + c\end{aligned}$$

24. Evaluate $\int \frac{3x^2 - x + 1}{x^3 - x^2} dx$.

Solution: We can write

$$\begin{aligned}\frac{3x^2 - x + 1}{x^3 - x^2} &= \frac{3x^2 - x + 1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} \\ &= \frac{0}{x} + \frac{-1}{x^2} + \frac{3}{x-1}\end{aligned}$$

$$\begin{aligned}\text{because } 3x^2 - x + 1 &= Ax(x-1) + B(x-1) + Cx^2 \Rightarrow A + C = 3, -A + B = -1, C = 3 \\ &\Rightarrow A = 0, B = -1, C = 3\end{aligned}$$

Therefore,

$$\begin{aligned}\int \frac{3x^2 - x + 1}{x^3 - x^2} dx &= -\int \frac{1}{x^2} dx + 3 \int \frac{1}{x-1} dx \\ &= \frac{1}{x} + 3 \ln|x-1| + c\end{aligned}$$

27. Evaluate $\int \frac{x^2}{(x+1)^3} dx$.

Solution: We can write

$$\begin{aligned}\frac{x^2}{(x+1)^3} &= \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} \\ &= \frac{1}{x+1} + \frac{-2}{(x+1)^2} + \frac{1}{(x+1)^3}\end{aligned}$$

$$\begin{aligned}\text{because } x^2 &= A(x+1)^2 + B(x+1) + C \text{ and if } x = -1, \text{ then} \\ C &= 1, A = 1, 2A + B = 0 \Rightarrow A = 1, B = -2, C = 1\end{aligned}$$

Therefore,

$$\begin{aligned}\int \frac{x^2}{(x+1)^3} dx &= \int \frac{1}{x+1} dx - 2 \int \frac{1}{(x+1)^2} dx + \int \frac{1}{(x+1)^3} dx \\ &= \ln|x+1| + \frac{2}{x+1} - \frac{1}{2(x+1)^2} + c\end{aligned}$$

30. Evaluate $\int \frac{dx}{x^3 + 2x}$.

Solution: We can write

$$\frac{1}{x^3 + 2x} = \frac{1}{x(x^2 + 2)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 2}$$

$$\Rightarrow 1 = A(x^2 + 2) + (Bx + C)x$$

If $x=0$, then we get $A = \frac{1}{2}$, $A + B = 0$, $C = 0 \Rightarrow A = 1/2$, $B = -1/2$, $C = 0$

Therefore, we get

$$\int \frac{dx}{x^3 + 2x} = \frac{1}{2} \int \frac{1}{x} dx - \frac{1}{2} \int \frac{x}{x^2 + 2} dx$$

$$= \frac{1}{2} \ln|x| - \frac{1}{4} \ln|x^2 + 2| + c$$

32. Evaluate $\int \frac{x^3 + x^2 + x + 2}{(x^2 + 1)(x^2 + 2)} dx$.

Solution: We can write

$$\frac{x^3 + x^2 + x + 2}{(x^2 + 1)(x^2 + 2)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 2}$$

$$\Rightarrow x^3 + x^2 + x + 2 = (Ax + B)(x^2 + 2) + (Cx + D)(x^2 + 1)$$

$$\Rightarrow A + C = 1, B + D = 1, 2A + C = 1, 2B + D = 2$$

$$\Rightarrow A = 0, B = 1, C = 1, D = 0$$

Therefore, we get

$$\int \frac{x^3 + x^2 + x + 2}{(x^2 + 1)(x^2 + 2)} dx = \int \frac{1}{x^2 + 1} dx + \int \frac{x}{x^2 + 2} dx$$

$$= \tan^{-1} x + \frac{1}{2} \ln|x^2 + 2| + c$$

34. Evaluate $\int \frac{x^4 + 6x^3 + 10x^2 + x}{x^2 + 6x + 10} dx$.

Solution: We can write by Long Division

$$\frac{x^4 + 6x^3 + 10x^2 + x}{x^2 + 6x + 10} = x^2 + \frac{x}{x^2 + 6x + 10} = x^2 + \frac{x}{(x+3)^2 + 1}$$

Let $x+3 = u$. Then $\frac{x}{(x+3)^2 + 1} = \frac{u-3}{u^2 + 1} = \frac{u}{u^2 + 1} - \frac{3}{u^2 + 1}$

Hence, we get

$$\int \frac{x^4 + 6x^3 + 10x^2 + x}{x^2 + 6x + 10} dx = \int x^2 dx + \int \frac{u}{u^2 + 1} du - 3 \int \frac{1}{u^2 + 1} du$$

$$= \frac{x^3}{3} + \frac{1}{2} \ln|u^2 + 1| - 3 \tan^{-1} u + c$$

$$= \frac{x^3}{3} + \frac{1}{2} \ln|(x+3)^2 + 1| - 3 \tan^{-1}(x+3) + c$$